

SAMPLE PAPER 1

Solve any one of the following Problems.

Question 1

A **Digit Permutation Cipher** is a simple form of number encryption where the digits of a number are rearranged based on a given key where ($1 \leq \text{key} \leq \text{size of the number}$). The key is a sequence of integers that defines the new positions of the digits.

Example: If number = 2613 and key = 4213, then the encrypted number will be 1632 by positioning the first digit to the 4th place, second digit to the second place, third digit to the first place and the fourth digit to the third place as per the key given.

Write a program to enter a number and a permutation key (a sequence of digits which is greater than 0 and less than or equal to the size of the number). The program should encrypt the number by permuting its digits according to the key. The number of digits in the key must match the number of digits in the number to be encrypted.

Test your program with the following data and some random data:

Example 1

INPUT: Number: 12345

Key: 31524

OUTPUT: The encrypted number is 24153

Example 2

INPUT: Number: 9876

Key: 4132

OUTPUT: The encrypted number is 8679

Example 3

INPUT: Number: 5239

Key: 4765

OUTPUT: INVALID KEY DIGITS

Example 4

INPUT: Number: 123

Key: 2134

OUTPUT: INVALID KEY SIZE

Question 2

Write a program to declare a single-dimensional array A [] of size L, where L is an integer greater than or equal to 3 and less than or equal to 50. Allow the user to input integers into this array. Display an appropriate error message for an invalid input.

Perform the following tasks on the array:

- (a) Display the array.
- (b) Find and print all equilibrium indices in increasing order. An index i (0-based) is called an equilibrium index if the sum of elements on its left equals the sum of elements on its right.
- (c) If no equilibrium index exists, display an appropriate message.

Test your program with the following data and some random data:

Example 1

INPUT: L = 7
Array Elements: 2, 3, -1, 8, 4, -2, 2
OUTPUT: Array: 2, 3, -1, 8, 4, -2, 2
Equilibrium Indices: 3

Example 2

INPUT: L = 5
Array Elements: 0, -7, 3, -2, 6
OUTPUT: Array: 0, -7, 3, -2, 6
Equilibrium Indices: 0

Example 3

INPUT: L = 6
Array Elements: 1, 2, 3, 4, 5, 6
OUTPUT: Array: 1, 2, 3, 4, 5, 6
Equilibrium Indices: NIL

Example 4

INPUT: L = 7
Array Elements: -7, 1, 5, 2, -4, 3, 0
OUTPUT: Array: -7, 1, 5, 2, -4, 3, 0
Equilibrium Indices: 3, 6

Example 5

INPUT: L = 2
OUTPUT: INVALID INPUT

Question 3

Write a program to accept a sentence which may be terminated by either '.', '?' or '!' only. The words may be separated by a single blank space and should be case-insensitive.

Perform the following tasks:

- (a) If the sentence is a pangram (a sentence or a phrase that uses every letter of the alphabet at least once), then print PANGRAM.
- (b) If it misses exactly one letter of the alphabet, print PANGRAMMATIC LIPOGRAM and the missing letter.
- (c) Else print NEITHER and list all missing letters in alphabetical order.

Test your program for the following data and some random data:

Example 1

INPUT: The quick brown fox jumps over the lazy dog!

OUTPUT: IT IS A PANGRAM

Example 2

INPUT: A quick movement of the enemy will jeopardize.

OUTPUT: PANGRAMMATIC LIPOGRAM

MISSING: x

Example 3

INPUT: Hello world.

OUTPUT: NEITHER

MISSING: a b c f g i j k m n p q s t u v x y z

Example 4

INPUT: Alas! it failed #

OUTPUT: INVALID INPUT

SAMPLE PAPER 2

Solve **any one** of the following Problems.

Question 1

A distinct-prime-digit integer is a positive integer (without leading zeros) in which all digits are prime numbers and no digit is repeated. The prime digits are 2, 3, 5, 7. For example, 2, 37, 253 are distinct-prime-digit integers, whereas 33, 252, 29 are not.

Given two positive integers m and n , where $m < n$, write a program to determine how many distinct-prime-digit integers are there in the range between m and n (both inclusive) and output them.

Test your program for the following data and some random data:

Example 1

INPUT: $m = 20$
 $n = 60$

OUTPUT: THE DISTINCT-PRIME-DIGIT INTEGERS ARE:
 23, 25, 27, 32, 35, 37, 52, 53, 57
 FREQUENCY OF DISTINCT-PRIME-DIGIT INTEGERS IS: 9

Example 2

INPUT: $m = 70$
 $n = 120$

OUTPUT: THE DISTINCT-PRIME-DIGIT INTEGERS ARE:
 72, 73, 75, 77
 FREQUENCY OF DISTINCT-PRIME-DIGIT INTEGERS IS: 4

Example 3

INPUT: $m = 100$
 $n = 180$

OUTPUT: THE DISTINCT-PRIME-DIGIT INTEGERS ARE: NIL
 FREQUENCY OF DISTINCT-PRIME-DIGIT INTEGERS IS: 0

Example 4

INPUT: $m = 200$
 $n = 150$

OUTPUT: INVALID INPUT

Question 2

Write a program to declare a single-dimensional array `arr[ ]` of size `N`, where `N` is an integer greater than 2 and less than 10. Allow the user to input positive integers into this array. Display an appropriate error message for an invalid input.

Rearrange the array in zig-zag order in the form $arr0 \leq arr1 \geq arr2 \leq arr3 \geq arr4 \dots$ using in-place swaps only.

Example: Input: [4, 3, 7, 8, 6, 2, 1]

Output: [3, 7, 4, 8, 2, 6, 1] after rearranging in zig-zag order.

Perform the following tasks on the array:

- (a) Display original array.
- (b) Transform to zig-zag order.
- (c) Display transformed array.

Test your program with the following data and some random data:

Example 1

INPUT: `N=5`
 Array Elements: 20, 5, 25, 30, 18

OUTPUT: Original Array: 20, 5, 25, 30, 18
 Zig-zag Array: 5, 25, 20, 30, 18

Example 2

INPUT: `N=8`
 Array Elements: 100, 50, 175, 25, 12, 150, 17, 200

OUTPUT: Original Array: 100, 50, 175, 25, 12, 150, 17, 200
 Zig-zag Array: 50, 175, 100, 150, 12, 25, 17, 200

Example 3

INPUT: `N=6`
 Array Elements: 21, 4, 34, -2, 9 10

OUTPUT: INVALID INPUT

Example 4

INPUT: `N=4`

OUTPUT: INVALID INPUT

Question 3

Write a program to accept a sentence which may be terminated by either '.', '?' or '!' only. The words may be separated by a single blank space and should be case-insensitive.

Perform the following tasks:

(a) Check if the sentence is a Palindrome Sentence.

[A sentence is a Palindrome Sentence if, after removing spaces and punctuation, the letters read the same forward and backward.]

Example: " Never odd or even "

(b) Display the first-occurring most frequent word in the sentence (in lower-case).

If there is a tie, choose the word that appears first in the sentence and if no words are repeated then print NONE.

Test your program for the following data and some random data:

Example 1

INPUT: No lemon no melon.

OUTPUT: IT IS A PALINDROME SENTENCE

MOST FREQUENT WORD: no

Example 2

INPUT: Was it a car or a cat I saw?

OUTPUT: IT IS A PALINDROME SENTENCE

MOST FREQUENT WORD: a

Example 3

INPUT: It is a rainy day!

OUTPUT: IT IS NOT A PALINDROME SENTENCE

MOST FREQUENT WORD: NONE

Example 4

INPUT: Always be careful#

OUTPUT: INVALID INPUT

SAMPLE PAPER 3

Solve any one of the following Problems.

Question 1

The **12-hour clock format** (A.M./P.M.) is a time convention in which the 24 hours of the day are divided into two periods. The **24-hour clock format** is the international standard.

Example: If time = 9:35 PM, then output will be 21:35

Write a program to accept a time in 12-hour clock format (HH:MM AM/PM) and convert it to 24-hour clock format (HH:MM).

Test your program for the following data and some random data:

Example 1

INPUT: 11:30 PM

OUTPUT: 23:30

Example 2

INPUT: 12:00 AM

OUTPUT: 00:00

Example 3

INPUT: 1:45 AM

OUTPUT: 01:45

Example 4

INPUT: 12:00 PM

OUTPUT: 12:00

Example 5

INPUT: 13:30 AM

OUTPUT: INVALID INPUT

Question 2

Write a program to declare a matrix A[][] of order ($M \times N$) where 'M' is the number of rows and 'N' is the number of columns such that both M and N must be greater than 2 and less than 10. Allow the user to input positive integers into this matrix. Display appropriate error message for an invalid input.

Perform the following tasks on the matrix:

- (a) Display the input matrix
- (b) Compute the **sum of prime elements** for each row and column
- (c) Identify the **row index** with maximum prime-sum and the **column index** with maximum prime-sum.
- (d) Print both indices and sums; if no primes found in all, print a message.

Example: Input: M=3 & N=4 and array is:

4	5	6	7
8	11	10	13
1	2	3	4

then, Output: Row with max prime-sum: 1 (sum=24)
Column with max prime-sum: 3 (sum=20)

Test your program for the following data and some random data:

Example 1

INPUT: M = 4
N = 4
Array elements: 11,3,4,7,13,9,5,17,2,3,19,23,12,6,13,4

OUTPUT: Original array:

11	3	4	7
13	9	5	17
2	3	19	23
12	6	13	4

Row with max prime-sum: 2 (sum=47)
Column with max prime-sum: 3 (sum=47)

Example 2

INPUT: M = 3
N = 3
Array elements: 2, 6, 4, 11, 5, 9, 12, 7, 13

OUTPUT: Original array:

2	6	4
11	5	9
12	7	13

Row with max prime-sum: 2 (sum=20)
Column with max prime-sum: 0 and 3 (sum=13)

Example 3

INPUT: M = 3
N = 3
Array elements: 12, 11, -5, 9, 1, 7, 13, 2, 3

OUTPUT: INVALID IN PUT

Example 4

INPUT: M = 2
N = 3

OUTPUT: INVALID INPUT

Question 3

Accept a sentence terminated by '.', '?' or '!' **only**. Sentence may contain any of ()[]{} plus letters/spaces. It must contain no digits and no other symbols besides the final terminator. For any violation, display: INVALID INPUT.

Perform the following tasks:

- (a) Check if the brackets are **balanced and properly nested**.
- (b) Print the maximum nesting depth of brackets.
- (c) If unbalanced, print the index (1-based indexing) of the first error.
 - A mismatched/extra closing bracket is an error at that bracket's index.
 - If brackets are incomplete at the terminator, the first error is the index of the first bracket that cannot be matched (typically where the mismatch first occurs).

Test your program for the following data and some random data:

Example 1

INPUT: This (is [very {deep}]) indeed!

OUTPUT: BALANCED: YES
MAX DEPTH: 3

Example 2

INPUT: Simple () test.

OUTPUT: BALANCED: YES
MAX DEPTH: 1

Example 3

INPUT: Hello world.

OUTPUT: BALANCED: YES
MAX DEPTH: 0

Example 4

INPUT: We love (balanced [brackets])!

OUTPUT: BALANCED: NO
ERROR AT INDEX: 22

Example 5

INPUT: Hi #there!

OUTPUT: INVALID INPUT